Ozone US-all Metric Inventory and Top-2 Summary

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1 Source and reading guide

This note summarizes the following workbook:

`D:/Github/spatial-skew-t/code/analysis/ozone/US-all/output/us-all/tables/comparison_top2.updated.xlsx`

The goal is to keep the metric definitions, ranking logic, and top-2 model configurations in one manuscript-ready LaTeX file.

In the workbook, `rank_value` is always the quantity used to order the top-2 rows, whereas `score_value` is the raw metric itself. The detailed tables below collapse each `ranking_group` into one row with a winner and a runner-up. Because every top-2 entry in this workbook has `CMAQ=yes`, `CMAQ` is documented in the column glossary but omitted from the per-row configuration strings.

2 Metric inventory

3 High-level patterns

The global probabilistic-score block still leans toward Morris-baseline skew-t settings. In particular, CRPS is led by settings 35 and 11, and many Brier and Quantile-score winners at low-to-moderate quantile levels remain in the Morris lane.

Upper-tail scoring is more heterogeneous. AR2 wins Brier at $q = 0.95$ and $q = 0.995$, and it also takes Quantile-score wins at $q = 0.98$, 0.99, and 0.995. MRTS wins Brier at $q = 0.9$, Quantile score at $q = 0.8$ and 0.9, and the below-threshold Brier split at $q = 0.98$ and 0.99.

Classification metrics show a precision-recall trade-off. High-threshold `Other` skew-t settings, especially settings 18 and 14 with `thr=85`, dominate precision and specificity. By contrast, MRTS appears most clearly in F1/Recall at $q = 0.98$, whereas calibration diagnostics remain mostly Morris or AR2: AR2 wins coverage at 0.8 and 0.9, and Morris still dominates most PIT-based summaries.

Block	Metrics	Scenario dimension	Top-2 ranking rule
Probabilistic score	Brier, Quantile score, CRPS	Brier thresholds $q \in \{0, 0.1, \ldots, 0.995\}$, quantile-score levels $q \in \{0.1, \ldots, 0.995\}$, and one overall CRPS summary	`ranking_basis = rel_to_gaussian`; smaller `rank_value` is better.
Tail-aware split score	Brier split	$q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ crossed with `above_threshold`, `all`, and `below_threshold`	`ranking_basis = rel_to_gaussian`; smaller `rank_value` is better.
Classification	Accuracy, Precision, Recall, Specificity, F1	Event quantiles $q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ with probability cutoff 0.5	`ranking_basis = raw_score`; the workbook already sorts in the correct direction for each metric.
Calibration to target	Coverage, PIT mean, PIT variance	Coverage levels $0.8, 0.9, 0.95$ plus overall PIT summaries	`ranking_basis = abs_gap_to_target`; smaller absolute gap is better.
Raw uncertainty diagnostics	PIT KS, PIT uniformity MAE, PIT uniformity RMSE	Overall summaries	`ranking_basis = raw_score`; lower values are better.

Table 1: Metric blocks represented in `comparison_top2.updated.xlsx`.

Workbook column(s)	Interpretation	How it appears in this draft
`metric_family`, `metric`	Identify the metric block and the named metric.	Used as the sectioning logic and the first column of each detailed table.
`ranking_group`	Source-side identifier for a specific metric/scenario combination.	Suppressed in the detailed tables because `metric + scenario` is easier to read in the manuscript.
`target_level`, `threshold_value`, `target_value`	Define the nominal quantile/coverage level and the corresponding event or target.	Condensed into the manuscript-side *Scenario* column.
`prediction_cutoff`	Probability cutoff used by classification metrics.	Included in the *Scenario* column for classification rows.
`band_type`	Split used by the Brier decomposition: `above_threshold`, `all`, or `below_threshold`.	Included in the *Scenario* column for Brier-split rows.
`ranking_basis`	Explains whether the workbook ranks by relative score, raw score, or absolute target gap.	Explained in the inventory table and used to interpret `rank_value`.
`rank_value`	Quantity actually used to assign rank 1 and rank 2.	Reported directly for both winner and runner-up.
`score_value`, `score_se`	Raw metric value and its standard error.	Reported as *Score* ± *SE*.
`setting`	Reproducible model identifier in the ozone US-all workflow.	Encoded inside each winner/runner-up label as `S<setting>`.
`method`, `knots`, `thresh`, `TS`, `ar2`, `mrts`	Core model-configuration fields.	Printed inside the winner/runner-up labels.
`model_lane`	Broad model family: Gaussian reference, Morris baseline, AR2, MRTS, or another numeric variant.	Printed inside the winner/runner-up labels as `Gaussian`, `Morris`, `AR2`, `MRTS`, or `Other`.
`CMAQ`	Indicator of whether CMAQ covariates are included.	Documented here only; all top-2 rows in this workbook have `CMAQ=yes`.

Table 2: Important workbook columns for manuscript use.

4 Detailed top-2 tables

Each winner/runner-up cell is a compact label containing the setting id, the model lane, the sampling family, and the key tuning fields `K`, `thr`, `TS`, `AR2`, and `MRTS`. The detailed tables retain the workbook's top-2 ordering but replace the source-side `ranking_group` string with a more compact scenario label.

4.1 Probabilistic scores

Table 3: Top-2 probabilistic-score summaries

Metric	Level	1st	Score	2nd	Score
Brier score	0.9	MRTS / skew-t (K=1, thr=0, AR1, MRTS=10)	0.04467	Morris / skew-t (K=1, thr=0, AR1)	0.04470
Brier score	0.95	AR2 / skew-t (K=7, thr=50)	0.02664	AR2 / skew-t (K=10, thr=50)	0.02672
Brier score	0.98	Morris / skew-t (K=6, thr=50, AR1)	0.01257	Morris / skew-t (K=5, thr=50, AR1)	0.01257
Brier score	0.99	Morris / t (K=9, thr=75, AR1)	0.00688	Morris / skew-t (K=5, thr=50)	0.00690
Brier score	0.995	AR2 / t (K=15, thr=75, AR2=yes)	0.00384	Morris / t (K=9, thr=75, AR1)	0.00385

Table 4: Top-2 probabilistic-score summaries

Metric	Level	1st	Score	2nd	Score
Quantile score	0.9	MRTS / skew-t (K=1, thr=0, AR1, MRTS=5)	5.033	Morris / skew-t (K=5, thr=0)	5.033
Quantile score	0.95	Morris / skew-t (K=7, thr=50, AR1)	2.988	AR2 / skew-t (K=10, thr=0)	2.988
Quantile score	0.98	AR2 / skew-t (K=10, thr=0)	1.423	Morris / skew-t (K=9, thr=0)	1.424
Quantile score	0.99	AR2 / skew-t (K=6, thr=50)	0.797	Morris / skew-t (K=9, thr=50, AR1)	0.798
Quantile score	0.995	AR2 / skew-t (K=6, thr=50)	0.441	AR2 / skew-t (K=8, thr=50)	0.442

4.2 Uncertainty and calibration

Table 5: Top-2 uncertainty and calibration summaries

Metric	Level	1st	Score	2nd	Score
Coverage	0.8	AR2 / skew-t (K=9, thr=50)	0.8001	MRTS / t (K=1, thr=75, AR1, MRTS=15)	0.8016
Coverage	0.9	AR2 / skew-t (K=8, thr=0)	0.9003	Morris / t (K=9, thr=75, AR1)	0.8996
Coverage	0.95	Morris / skew-t (K=2, thr=0)	0.9499	Morris / skew-t (K=4, thr=0)	0.9499

5 Test robustness of number of samples

This section records how the top-ranked setting changes when the number of training and validation sites changes.

Exceedance thresholds (full dataset).

Quantile	Threshold
0.9000	73.2500
0.9500	79.7500
0.9800	88.5000
0.9900	94.7500
0.9950	100.5200

5.1 Use 300 sites to predict 100 sites (4216.09 sec)

Validation selection result (summary draw = 5000, top-4).

Quantile	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	204 (0.041160)	55 (0.041325)	205 (0.041356)	70 (0.041393)
0.9500	70 (0.019694)	41 (0.019701)	12 (0.019776)	39 (0.019819)
0.9800	204 (0.007674)	117 (0.007763)	67 (0.007807)	111 (0.007820)
0.9900	117 (0.003095)	15 (0.003146)	70 (0.003161)	61 (0.003180)
0.9950	15 (0.001272)	70 (0.001297)	61 (0.001301)	117 (0.001306)

5.2 Use 200 sites to predict 200 sites (2267.64 sec)

Validation selection result (summary draw = 5000, top-4).

Quantile	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	54 (0.050060)	105 (0.050220)	35 (0.050348)	215 (0.050382)
0.9500	117 (0.030306)	116 (0.030322)	114 (0.030328)	70 (0.030345)
0.9800	215 (0.015037)	205 (0.015049)	111 (0.015052)	58 (0.015067)
0.9900	58 (0.007714)	112 (0.007727)	216 (0.007740)	67 (0.007743)
0.9950	29 (0.004225)	35 (0.004227)	60 (0.004227)	116 (0.004251)

5.3 Use 400 sites to predict 400 sites (7829.68 sec)

Top-2 by quantile.

Quantile	Threshold	Rank 1 setting	Brier (Rank 1)	Rank 2 setting	Brier (Rank 2)
0.900	73.25	204	0.04466516	51	0.04469537
0.950	79.75	111	0.02664004	120	0.02672309
0.980	88.50	58	0.01256908	55	0.01257193
0.990	94.75	68	0.00687592	8	0.00690039
0.995	100.52	124	0.00384228	68	0.00385141

5.4 Remark

With the same MCMC iteration budget, changing the train/validation split to 300/100 does not yield fully consistent winners across all target quantiles. The practical conclusion is therefore to report this instability directly, rather than forcing a single setting to represent all tails.